

Comparison of Site Preparation and Revegetation Strategies Within a Sphagnum-dominated Peatland Following Removal of an Oil Well Pad

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ABSTRACT

Few guidelines exist for the effective revegetation of peatlands following the removal of in-situ oil and gas infrastructure. We conducted a manipulative field study in northeast Alberta, Canada, on a well pad undergoing removal and revegetation to test different management practices for facilitating vegetation establishment and recovery of a sphagnum-dominated peatland. A randomized block design was used to evaluate the effect of various revegetation practices, including augmenting natural recovery with the use of native transplants, acrotelm transfer from a similar intact donor peatland, and the use of variable surface microtopography. Although overall survival of transplants was similar between areas smoothed and left rough, areas that were rough had greater species richness under natural recovery, including trees, shrubs, and other perennial herbs. Moreover, survival and growth of woody transplants (*Picea mariana*—black spruce and *Ledum groenlandicum*—Labrador tea) were greater when planted within the top and middle microtopographic positions rather than micro-depressions. Survival of transplanted sedges (*Carex* spp.) was high at all topographic positions, but benefited the most in growth from planting in depressions. Contrary to expectations, no benefits of acrotelm application were found on vegetation recovery during the first two seasons, and even reduced the presence of some native vegetation. Although transplants directly contributed to revegetation, the recovering peatland remained highly dissimilar (> 90%) in composition relative to the neighboring peatland after two years.

Keywords: acrotelm, Canada, *Carex* spp., *Ledum groenlandicum*, microtopography, natural recovery, *Picea mariana*, transplants

🌿 Restoration Recap 🌿

- To test different management practices for facilitating vegetation establishment and recovery of a sphagnum-dominated peatland, we examined restoration with native transplants, acrotelm transfer from a similar intact donor peatland, and the use of variable surface microtopography
- Topographically rough surfaces tended to have greater species richness during the first two seasons.
- Woody species performance was greatest when installed above inundated areas, while sedges performed best in micro-depressions.
- Restoration activities that add microtopographic complexity may be an effective management strategy to mitigate ecological risk to transplants from a fluctuating water table.

Energy exploration and extraction (conventional oil and gas, oil sands) is ubiquitous in boreal regions of Alberta, Canada, producing a large network of production well pads, roads, and additional in-situ infrastructure (Schneider and Dyer 2006). Many of these features are

constructed within peatlands because approximately half of Alberta's boreal is comprised of this land type (Vitt et al. 1996), where they disrupt natural functions such as carbon sequestration, water filtration, and natural drainage (Graf 2009). Well pads and roads are typically constructed by placement of predominantly clay fill excavated from adjacent uplands on top of the peat surface with either timber corduroy or a fabric liner separating mineral soil from peat. Common practice for reclamation of such sites has been to leave the fill and convert the affected area to an upland

community by planting grasses, shrubs, and trees. Increasingly however, regulatory emphasis is on removing the fill and restoring peatland vegetation, hydrologic connectivity, and function (Alberta Energy Regulator, pers. comm.).

Removal of road bed fill can be problematic during reclamation depending on physical characteristics of underlying peat, depth of fill, and whether the peat was correctly loaded (Forestry Civil Engineering 2010) to achieve floatation of the pad or road on top of the peat (Osisko 2010). Options include the complete removal of fill or merely lowering fill elevation to near the water table, either of which may include removal or abandonment of geotextile liners and reinforcing materials, and the use of amendments for subsequent reclamation. Although complete fill removal may aid revegetation of organic substrates, this can be problematic in situations where large amounts of fill has either displaced peat or compressed it far below the water table. Subsequent removal of submerged fill results in a void filled with water (e.g., AMEC 2011). On the other hand, leaving fill in-situ confounds soil wetting for plant establishment and peatland development during annual fluctuations in the water table. Furthermore, chemistry of geologic material in boreal Alberta varies (Natural Regions Committee 2006) and such material used for construction can differ from or influence the peatland in which it is deposited (Alberta Environment and Parks 2015, Wood et al. 2015). For example, the presence of clay can physically impede water flow as well as elevate the cation exchange capacity of peat-based soils where contact occurs (Sithole and Guy 1985).

Vitt et al. (2011) addressed rewetting experimentally with a terraced surface on well pads leveled to the water table, and concluded that wetland plants could be established on mineral substrate provided correct elevations were achieved, though this may be difficult as it requires extensive data on seasonal and annual variation in the water table. Terracing to provide multiple elevations may help, but also produces large areas with conditions too wet or too dry for revegetation. Creating a rough or undulating surface within the target elevation range could address water table variations by ensuring there are always areas of suitably wetted substrate available regardless of water level fluctuations.

Appropriate plant establishment must match the hydrologic and ecologic functions of a peatland if it is to be sustainable (Price et al. 2010). Most peatlands in boreal Alberta were initiated by paludification, or development of peat over previously forested, grassed, or bare soils (Kuhry and Turunen 2006, Bloise 2007). Vitt et al. (2011) emulated paludification by introducing wetland pioneer species (*Carex aquatilis*, *Salix lutea*, *Larix laricina*) onto rewetted mineral soil remaining after leveling well pads to near the water table. While able to establish wetland plants, Vitt et al. (2011) were uncertain whether representative plant communities would develop in terms of species richness,

community structure, and carbon sequestration. Others have successfully re-established peatland vegetation on cutover or mined peatlands by transplanting the acrotelm (living) layer of peat (Cagampan and Waddington 2008a, 2008b) or introducing acrotelm material onto organic soils (Quinty and Rochefort 2003, Cobbaert et al. 2004, Graf and Rochefort 2008, 2010). It is unknown whether these techniques would be equally successful on mineral soil or peat-mineral mixes.

We examined a well-pad initially built in a sphagnum-dominated peatland in northeastern Alberta (classed as a “treed poor fen”, ca. Beckingham and Archibald 1996), from which clay fill had been almost entirely removed, with the remainder churned under the underlying organic soil to facilitate revegetation. We then applied a replicated, randomized matrix of site preparation and revegetation treatments, including: surface roughness; transplants of graminoid, shrub, and tree species; the introduction of acrotelm material; and untreated control sites. Our first hypothesis was that plots receiving transplants would demonstrate greater recovery when planted into areas with a greater diversity of niche microsites and receiving peat application, and therefore be more similar to the adjacent natural peatland. Our second hypothesis was that introducing acrotelm would increase species richness and diversity, particularly within naturally recovering plots.

Methods

Study Sites

The study was conducted on an abandoned Canadian Natural Resources Ltd. (CNRL) well-pad situated 40 km northwest of Cold Lake, Alberta (54°24'18" N; 110°16'46" W). The site is within the Eastern Alberta Plains physiographic region near the border of the Dry and Central Mixedwood Subregions (Natural Regions Committee 2006). Long-term annual precipitation at Cold Lake averages 427 mm, with 80% occurring from April to September, with an average temperature of 1.7°C (Environment Canada, unpub. data). Peatland soils are organic in nature, often Fibric Mesisols.

The study site initially was a 140 × 130 m well-pad constructed in 1997 within a treed poor Sphagnum peatland using fill from a nearby borrow pit. The area was cleared by logging larger trees and pushing small trees down with a dozer, then 1.5 m of non-saline (0.76 dS/m; Smith and Doran 1996) clay/sandy-clay fill was deposited. Most fill was subsequently removed in November 2010 with a track-hoe, although a thin layer (approximately 10 cm thick) remained across much of the study site. The surface was then churned with a track-hoe in an attempt to bury the remaining fill and bring up organic soil from below, resulting in a mounded surface (up to 1 m relief) composed of a heterogeneous mosaic of interspersed patches of mineral soil and well-humified peat. Soil samples taken across

Table 1. Mean cover of the reference community adjacent to the Canadian Natural Resources Ltd (CNRL) well pad, Alberta, Canada, by growth form and species.

Growth Form	Species	Cover (%)
Trees and Shrubs	<i>Betula pumila</i> var. <i>glandulifera</i>	1.0
	<i>Ledum groenlandicum</i>	40.0
	<i>Oxycoccus microcarpus</i>	12.5
	<i>Picea mariana</i>	15.3
	<i>Ribes hudsonianum</i>	1.9
	<i>Salix arbusculoides</i>	13.8
	<i>Vaccinium myrtilloides</i>	1.3
	<i>Vaccinium vitis-idaea</i> ssp. <i>minus</i>	25.6
Sub-shrubs and Forbs	<i>Drosera rotundifolia</i> var. <i>comosa</i>	0.6
	<i>Fragaria vesca</i> ssp. <i>americana</i>	1.3
	<i>Pyrola secunda</i> – <i>Orthilia secunda</i>	0.5
	<i>Rubus chamaemorus</i>	24.6
	<i>Rubus pubescens</i>	0.6
	<i>Smilacina trifolia</i>	20.4
	<i>Thalictrum venulosum</i>	0.5
	<i>Equisetum fluviatile</i>	0.5
Horsetails	<i>Equisetum scirpoides</i>	0.5
	<i>Eleocharis acicularis</i>	0.3
Sedges and Rushes	<i>Carex tenuiflora</i>	4.3
	<i>Carex utriculata</i>	3.8
	<i>Phragmites australis</i>	15.6
Grasses	<i>Aulacomnium palustre</i>	1.3
Mosses	<i>Hylocomium splendens</i>	1.5
	<i>Ptilium crista-castrensis</i>	0.6
	<i>Polytrichum strictum</i>	4.4
	<i>Sphagnum girgensohnii</i>	48.4
	<i>Sphagnum warnstorffii</i>	26.0

the study area and assessed for soil chemistry revealed a mean pH of 7.27, electrical conductivity of 421 $\mu\text{S}/\text{cm}$, and concentrations of calcium, sodium, magnesium and potassium of 41.4, 42.9, 21.4, and 4.5 mg/L. Total nitrogen and phosphorus levels were 1.85 and 0.04%, respectively. The adjacent peatland community was dominated by a diverse mix of species, including *Picea mariana* (black spruce), *Salix* sp. (dwarf willow), *Betula glandulosa* (bog birch), and *Ledum groenlandicum* (Labrador tea), along with a variety of sedges (*Carex* spp.), forbs, feather mosses, and peat mosses (see Table 1 for complete listing).

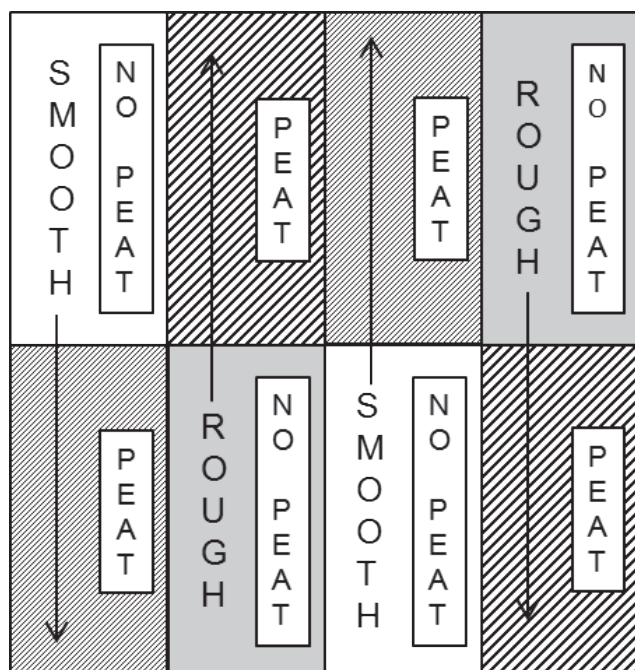
Experimental Design and Treatments

The revegetation treatments of surface roughness (i.e., site preparation), acrotelm application, and transplanting (vs. natural recovery) were tested with a fully randomized split-split plot design replicated four times in blocks measuring 40 m $\times$ 40 m in size and separated by working alleys. Blocks contained 4 macroplots, each 10 m $\times$ 40 m in size, two of which were randomly assigned for smoothing with a bulldozer in late March 2012, while the other two were left rough (Figure 1). Several equipment passes resulted in minimal relief (i.e., < 15 cm) in smoothed macroplots, while rough plots had prominent relief up to 1 m, creating a variety of microsites for vegetation establishment.

Macroplots were further divided into two 10 $\times$ 20 m mesoplots, one of which was randomly assigned to acrotelm application and the other represented natural recovery (i.e., no inoculation). While acrotelm transfer can involve the intact transfer of a thick peat layer (Karofeld et al. 2015), a modified version was described by Quinty and Rochefort (2003) that employed a rotary tiller to collect donor peat and a manure spreader for application. We were similarly interested in testing equipment commonly available in oilfield construction and reclamation for acrotelm transfer, and therefore used a modified version of the latter. Donor acrotelm was collected 20 km north from a similarly classified treed poor fen (Golder Associates 2011) undergoing development. Acrotelm was skimmed off the surface to 15 cm depth, loaded, transported to the CNRL study site and stockpiled, then applied within 7 days using a track-hoe with a “thumb” attachment, which ‘sprinkled’ acrotelm material from the bucket. Approximately 4 m³ of mixed peat was applied to each plot, resulting in a depth of 1–2 cm. Track-hoe traffic was applied to all plots to ensure consistent traffic and compaction effects. To avoid excessive soil desiccation from sun and wind, we applied a thin layer of wheat straw (1–2 cm) to all plots.

Random samples of donor acrotelm were removed at the time of transport and later tested for live propagules to

A



B

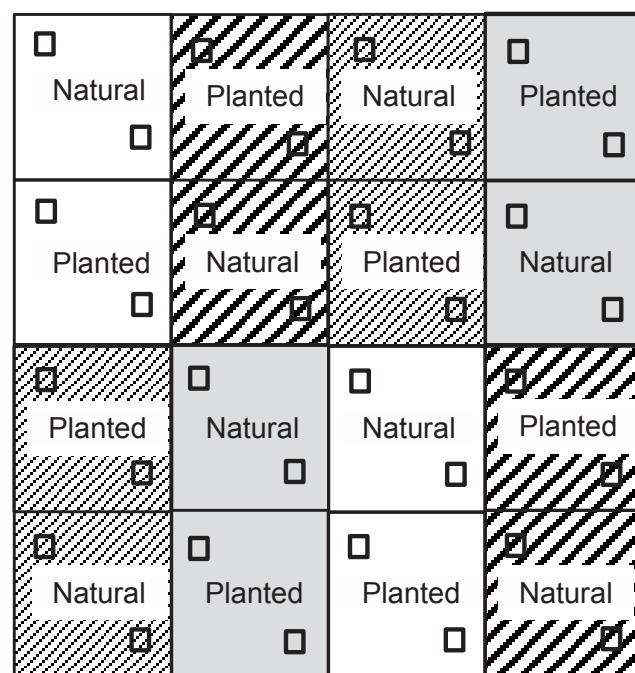


Figure 1. Sample layout of A) mesoplots (10 × 20 m) receiving various roughness and acrotelm amendment treatments within one study block, and B) the additional natural recovery and transplant split-plot microplots superimposed on top of the treatments described in A. Also shown are the quadrats (2 × 2 m) used for subsampling.

assess vegetation regeneration potential. Acrotelm samples were transported to the greenhouse at the Alberta Innovates Technology Futures station in Vegreville, Alberta, and grown out. Donor acrotelm material (1.95 L) was spread out in each of 40 trays at a depth of 2 cm over 2 cm of mineral potting mix during May 2012 and inspected biweekly for four months. Potting mix contained a mineral fraction to improve the cation exchange capacity and better mimic field conditions, as well as maximize opportunities for germination and identification (Verhagen 2004). All trays were watered weekly, and assessed for species richness and composition of vascular plants. In addition, monthly cover estimates of mosses and lichens were recorded. The number of vascular plants and cover of moss and lichens emerging from the acrotelm in greenhouse samples were recorded. All emergence data were summarized for each of the 40 trays, and used to report mean density of vascular plants and cover of bryophytes, along with variance among trays.

A second set of microplots (10 m × 10 m) was established within each mesoplot in the field to compare unassisted vegetation recovery (with or without acrotelm) to assisted recovery using transplants (Figure 1). Permanent quadrats (2 m × 2 m) within the transplant treatments were planted with *P. mariana*, *L. groenlandicum*, and sedge ramets removed from the adjacent peatland. While sedges were difficult to identify during transplanting, subsequent monitoring revealed a mix of *Carex aquatilis* (~ 30%) and *C. utriculata* (~ 70%). Quadrats in all other microplots were assessed without the addition of transplants.

Transplants were installed in late May of 2012. Within smoothed microplots, three plants of each species were planted per quadrat, but at one micro-topographic location—approximately mesic conditions (n = 96 plants/species). In rough microplots, quadrats were planted with each species at 3 micro-topographic locations, including: 1) drier mound tops; 2) mesic mid slopes; and 3) hydric micro-depressions about 1 cm above water, with three plants installed at each location in each quadrat (n = 288 plants/species). This stratification allowed testing of three elevations, as well as overall comparison with smoothed treatments.

Vegetation Assessment

Transplant survival was evaluated in early August of 2012 (i.e., ‘first’ growing season) and 2013 (‘second’ season). *Picea mariana* growth was determined in the second season as the height difference from May to August. To assess *P. mariana* height growth, final height was relativized to initial tree height in May 2013 to adjust for variation in starting tree size (Garnier and Navas 2012), resulting in a relative height ratio (RHR), shown by Equation 1:

$$\text{RHR} = \text{final height}/\text{initial height}$$

Sampling of vegetation and bryophyte recovery within microplots was confined to two permanently marked

quadrats, each 2 m × 2 m in size. Composition was visually assessed in August of the second season as percent cover by species, and in transplanted plots, for the cover of sedges. All nomenclature of vascular plants followed Moss (1983), while bryophytes followed Mackinnon et al. (1995). Two 20 m long control transects, situated 50 m away from and parallel to the edge of the well-pad, were used to characterize the undisturbed natural peatland surrounding the study site. Eight plots were systematically distributed every 2 m along each transect using a nested design: 1 m × 1 m size quadrats for small trees and shrubs and 0.5 m × 0.5 m size quadrats for herbs (including sedges), mosses, and lichens. Diversity was calculated for all plots using Equation 2:

$$H' = -\sum [\ln(p_i) \times (p_i)], \text{ where}$$

p_i is the proportion of total cover comprised of species 'i', and $\ln$ is the natural log of the same, and $\sum$ is the sum of values for all species within the plot.

Data Analysis

Survival, richness and diversity, *P. mariana* RHR, and transplant cover data were subsequently assessed with an analysis of variance (ANOVA) for each year using mixed models in Statistical Analysis Software v 9.4 (SAS Institute Inc., Cary, North Carolina), for a split-split plot design. Surface treatment, peat fragment application, and their interaction were tested as fixed factors, with block and replicate within block as random. A separate analysis was performed on planting position and its interaction with acrotelm application using only the rough surface plots to evaluate the contribution of microtopography to transplant survival and community responses within these treatments.

Community composition sampled in the second growing season within the natural recovery microplots (i.e., those lacking transplants) was examined using a permutation-based multiple analysis of variance (PerMANOVA) to determine whether composition differed among the surface roughness and acrotelm treatments (Anderson 2001). PerMANOVA was performed using the Sorensen distance measure, with all multivariate analyses done in PC-Ord v.5 (MjM Software Design, Gleneden Beach, Oregon). Significance was based on the proportion of randomized trials with an indicator cover greater than or equal to the observed cover (McCune and Grace 2002) based on Equation 3:

$$p = (1 + \text{number of runs} \geq \text{observed}) / (1 + \text{number of randomized runs})$$

An indicator species analysis (ISA) was used to test the correlation between individual species and fixed treatment classes (surface and peat fragment treatments) using 4999 permutations (McCune and Grace 2002).

To quantify overall vegetation recovery in relation to the revegetation treatments, composition in each microplot was compared to that from transects in the neighboring

peatland. As a first step, similarity between each of the 128 microplots (64 natural and 64 transplanted) was compared to composition in the peatland using the identical sample area (i.e., 2 m²). Similarity in species richness was then calculated using Jaccard's coefficient (Jaccard 1912) as per Equation 4:

$$S_j = a / (a + b + c)$$

Where a = number of species common to (shared by) quadrats, b = number of species unique to the first quadrat, c = number of species unique to the second quadrat. Similarity data were analyzed with mixed model ANOVAs to evaluate direct effects of surface treatment, acrotelm addition, transplanting, and their interactions.

Results

Acrotelm Composition

Emergent vegetation and bryophyte cover grown from acrotelm in the greenhouse increased over the first 8 weeks, then remained relatively constant. Small cranberries, mosses, and sedges were the most common and abundant species (Table 2). Despite a low frequency of appearance, trees such as *Populus tremuloides* (trembling aspen) had a high density in those trays where they occurred. The biennial species *Beckmannia syzigachne* (slough grass) reached abundant levels, both in density and frequency, after only 4 weeks.

Transplant Survival and Growth

Sedge survival in the field remained high in each of the first two years (98.2 and 85.1%, respectively), while both *P. mariana* and *L. groenlandicum* survival declined sharply from more than 50% the first year to less than 20% one year later for both woody species. Overall, final survival of transplants, including *P. mariana* (rough = 11.5%; smooth = 13.5%), *L. groenlandicum* (rough = 13.6%; smooth = 17.7%), and sedges (rough = 83.7%; smooth = 86.5%), did not differ significantly with surface roughness in August of the second growing season. We therefore conclude that surface roughness had little overall effect on transplant survival leading us to reject our hypothesis predicting greater transplant establishment within roughened surfaces. Similarly, acrotelm application had no effect on the aggregate survival of *P. mariana* (with acrotelm = 12.2%; without acrotelm = 12.8%), *L. groenlandicum* (with acrotelm = 13.9%; without acrotelm = 17.4%), or sedges (with acrotelm = 84.4%; without acrotelm = 85.8%).

Among the contrasting planting positions within rough surface plots, *P. mariana* survival was significantly affected by a position by acrotelm interaction during the first growing season (ANOVA; $F_{2,21} = 3.84$, $p = 0.049$). The greatest *P. mariana* survival in rough plots occurred at the upper position of acrotelm-treated plots, which differed

Table 2. Mean ($\pm$ SD) vascular and non-vascular plant characteristics emerging from donor acrotelm material grown within trays after 12 weeks.

Growth Form	Species	Mean Density (plants tray ⁻¹)	Mean Cover (% tray ⁻¹)	Total Frequency (% of trays)
Trees	<i>Picea mariana</i>	0.4 ($\pm$ 0.5)		15.0
	<i>Populus tremuloides</i>	2.5 ($\pm$ 5.3)		52.5
Shrubs	<i>Betula pumila</i>	0.8 ($\pm$ 0.3)		20.0
	<i>Ledum groenlandicum</i>	0.5 ($\pm$ 1.3)		17.5
	<i>Vaccinium myrtilloides</i>	2.3 ($\pm$ 4.7)		55.0
	<i>Vaccinium oxycoccos</i>	8.6 ($\pm$ 7.9)		92.5
	<i>Fragaria vesca</i>	0.6 ($\pm$ 2.5)		5.0
Forbs	<i>Equisetum arvense</i>	0.2 ($\pm$ 0.9)		25.0
	<i>Carex utriculata</i>	9.5 ($\pm$ 5.2)		100.0
Grass	<i>Beckmannia syzigachne</i>	1.1 ($\pm$ 2.5)		27.5
Mosses	<i>Polytrichum strictum</i>		6.8 ($\pm$ 14.4)	70.0
	<i>Sphagnum girgensohnii</i>		65.5 ($\pm$ 23.7)	100.0
Lichen	<i>Cladina rangiferina</i>		0.9 ($\pm$ 6.3)	27.5

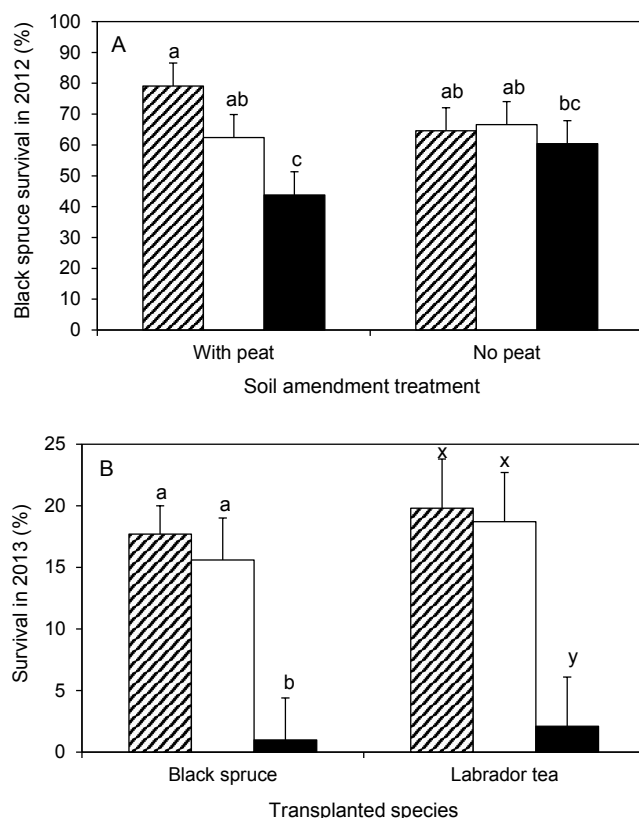


Figure 2. Comparison of mean ($\pm$ SE) transplant survival of A) *Picea mariana* during the first growing season relative to acrotelm application and microtopographic planting position within the rough surface treatments, and B) *P. mariana* and *Ledum groenlandicum* during the second growing season across different micro-topographic positions. Planting locations are shown for the top (cross-hatch), middle (white) and bottom (black) positions. Means with different letters are significantly different at $p < 0.05$.

significantly from the bottom position of the same (Figure 2A). In contrast, *P. mariana* survival did not differ significantly across positions in the absence of acrotelm. Additionally, significant position effects, but not acrotelm effects, remained evident for each woody transplant species in the second growing season (ANOVA for position; *P. mariana*: $F_{2,16} = 10.99$, $p = 0.0002$; *Ledum groenlandicum*: $F_{2,14} = 13.0$, $p < 0.0001$). *Picea mariana* and *L. groenlandicum* survival both remained significantly lower at the bottom position compared to the top and middle micro-topographic locations (Figure 2B). *Ledum groenlandicum* was additionally affected significantly by acrotelm application during the first growing season (ANOVA; $F_{1,14} = 4.18$, $p = 0.048$), with survival of this species lower in areas receiving acrotelm (50%) compared to areas without acrotelm (60.4%). Sedge survival was not affected significantly by any treatment in the second season and remained uniformly high, ranging from 84.4 to 87.5% among the three planting positions.

Values of RHR above 1 indicate net increases in *P. mariana* growth over the second growing season. *Picea mariana* RHR ranged from 0.97 to 1.15 among plots, and was significantly affected by planting position (ANOVA; $F_{2,11} = 4.26$, $p = 0.033$), but not surface or acrotelm treatment. Trees at the bottom position of rough plots (1.012) had significantly lower RHR compared to those trees alive within the top (1.053) and middle (1.074) locations, with the latter improving up to 6.2% in height growth.

At the end of the final year, the cover of *L. groenlandicum* and sedge transplants were significantly affected by planting position (ANOVA; *Ledum groenlandicum*: $F_{2,6} = 13.9$, $p < 0.0001$; sedges: $F_{2,14} = 10.2$, $p = 0.0003$), but not acrotelm application or their interaction. Cover of planted *L. groenlandicum* was significantly lower ($p < 0.05$) at the bottom position (0.1%) compared to the top (0.8%) and middle (1.2%) positions. In contrast, sedge transplant cover remained significantly lower ($p < 0.05$) within the

Table 3. Summary of indicator species analysis (ISA) of local plant species found on the naturally recovering plots in the second establishment year (2013) at the well-site. Only species with significant indicator values (IV) are listed (minimum $p < 0.10$).

Growth Form	Longevity	Plant species	Roughness Treatment			Acrotelm Treatment		
			Preferred Class	IV	p -value	Preferred Class	IV	p -value
Sedge	Perennial	<i>Carex aquatilis</i>	Smooth	41.9	0.017	—	—	—
Forb	Annual/Biennial	<i>Senecio congestus</i>	Smooth	63.1	0.038	—	—	—
Forb	Perennial	<i>Potentilla palustris</i>	Smooth	30.1	0.058	—	—	—
Shrub	Perennial	<i>Salix candida</i>	Rough	65.5	0.010	—	—	—
Shrub	Perennial	<i>Salix lucida</i>	Rough	46.3	0.042	—	—	—
Tree	Perennial	<i>Populus tremuloides</i>	Rough	63.9	0.010	No acrotelm	58.2	0.032
Tree	Perennial	<i>Populus balsamifera</i>	Rough	43.0	0.022	No acrotelm	40.5	0.044
Moss		<i>Ceratodon purpureus</i>	Rough	72.9	0.002	—	—	—
Moss		<i>Polytrichum juniperinum</i>	Rough	56.9	0.005	No acrotelm	47.1	0.064
Shrub	Perennial	<i>Ledum groenlandicum</i>	—	—	—	No acrotelm	31.5	0.053
Liverwort		<i>Marchantia polymorpha</i>	—	—	—	No acrotelm	69.4	0.001
Shrub	Perennial	<i>Salix exigua</i>	—	—	—	No acrotelm	35.7	0.071

top microtopographic position (2.8%) compared to the middle (3.9%) and bottom (4.0%) positions.

Composition Responses

Final species richness in the second growing season within the (non-planted) natural recovery plots was significantly affected by surface roughness (ANOVA; $F_{1,12} = 5.82$, $p = 0.025$), but not acrotelm application or their interaction. Although richness was highly variable among naturally revegetated plots, ranging from 1 to 29 species plot⁻¹, rough plots had more species ($n = 20.3 \pm 2.1$) on average than smoothed plots ($n = 16.6 \pm 2.1$). Shannon's diversity under natural recovery was significantly affected by surface roughness (ANOVA; $F_{1,11} = 2.94$, $p = 0.10$): diversity tended to be lower on smooth plots (1.994 ± 0.158) compared with those having rough microtopography (2.258 ± 0.158). Last, overall species diversity within plots receiving acrotelm (2.046 ± 0.158) remained similar to that in plots not receiving acrotelm (2.206 ± 0.158).

The collective species composition of treatment plots differed significantly in response to surface roughness (PerMANOVA; $F_{1,31} = 1.85$; $p = 0.05$) but not acrotelm application. The corresponding ISA showed several trees, shrubs, and mosses were positively associated with the rough surface treatment (Table 3). In contrast, sedges and short-lived (annual and biennial) forbs were positively related to the smooth surface treatment. Several trees, shrubs, and bryophytes were positively associated with plots lacking acrotelm application (Table 3). No species were associated with areas receiving acrotelm.

Finally, indices of Jaccard's similarity between the adjacent peatland and plots exposed to the different treatments did not vary significantly in relation to any of the revegetation strategies tested, including surface roughness and acrotelm application. For all treatment combinations, species similarity with the neighboring peatland remained very low ($< 10\%$) after two years.

Discussion

Natural Revegetation in Relation to Microtopography

While the difference in naturally colonizing species richness and diversity observed between smooth and rough treatments may be confounded by the additional mechanical disturbance associated with the smoothing treatment during reclamation (McEvoy et al. 1993), we can not rule out the possibility that divergence in recovery was aided by a greater variety of niches on roughened plots. Mid- to late seral woody species (trees and shrubs), as well as mosses, were all associated with rough plots in this study. Moss association (*Ceratodon* spp. and *Polytrichum* spp.) with the rough plots may reflect the increased likelihood of at least a portion of the plot area being close to the water table, as water is critical for moss survival (Kangas et al. 2014). Similarly, association of trees and shrubs with rough plots may reflect the increased presence of better drained microsites (i.e., mound-tops) with soil conditions conducive to shrub and tree establishment and survival. *Populus tremuloides* (Aspen) and *Populus balsamifera* (balsam poplar) colonization from seed, and possibly that of *Salix* spp. (willow), is typically a slow process reliant on highly variable seed entry, with seedling germination and survival occurring infrequently for small seeded species like these (Yarie 1993). Greater establishment of these species on rough plots may also be more likely with an uneven surface due to improved airborne seed capture (Blood and Titus 2010), and by providing the small scale microsite heterogeneity effective for seed trapping and survival (Janousek and Folger 2013).

In theory, microsite variability provides a greater variety of niches necessary to host more species. The greater richness found on rough rather than smoothed plots suggests the former may have been beneficial for increasing diversity, and therefore resiliency during vegetation recovery. In

contrast, aquatic sedges and ruderal (disturbance-adapted) forbs were associated with smoothed plots in this investigation. Forbs such as *Senecio congestus* (marsh ragwort) represented early seral species likely colonizing smooth plots because of the mechanical disturbance associated with smoothing, which took place at the end of the reclamation process (McEvoy et al. 1993). *Potentilla palustris* (marsh cinquefoil), another early to mid-seral forb that commonly colonizes disturbed areas (Bastl et al. 2009), may have similarly taken advantage of the mechanical disturbance, whereas *Carex aquatilis* (water sedge) may have colonized smoothed plots based on their being ideally adapted to the 'intermediate' moisture conditions within this mid-slope position (Deng et al. 2013).

Transplants for Enhancing Peatland Revegetation

No differences in *P. mariana* and *L. groenlandicum* survival and growth were observed overall between the rough and smooth treatments, indicating a rough surface may not be necessary for the enhancement of transplanted tree and shrub establishment, provided sufficient 'mesic' microsites are available on a smooth surface. While survival and growth of transplanted woody species on smoothed treatments did not differ from that on rough treatments, we suspect this finding was because smoothed plots were sufficiently elevated above the water table to avoid standing water.

Factors contributing to the mortality of trees and shrubs due to flooding are well known and include reduced oxygen for root respiration (Blom and Voesenek 1996). Had there been greater fluctuation in the water table, greater mortality of transplanted trees and shrubs may have occurred on the smoothed plots, as well as at the mid-slope positions of rough treatments. This is borne out by differences in tree and shrub survival and growth observed among planting positions within the rough treatment. Woody transplant survival was poorest at the bottom position, likely due to high anaerobic conditions and commensurate reductions in root respiration (Jeglum 1974, Kozłowski 1997), as well as cooler soil temperatures (Kozłowski and Pallardy 1997). As a result, using surface roughness as a strategic means to reduce ecological risk and improve resistance to periodic and unpredictable flooding may still be an effective strategy for improving transplant survival where partial fill removal is used as a reclamation practice and very precise water level control is not possible. Long-term monitoring is warranted to more fully test the long-term effects of these treatments on transplant survival.

Overall, species richness and diversity were marginally greater with the introduction of transplanted species, apparently as a direct result of the additional species installed. However, effects of transplants on overall biodiversity were small, though additional changes in diversity are expected as transplants grow and expand their influence on their surroundings. These results suggest transplantation with

carefully selected species appears to be a useful strategy for increasing revegetation success. Transplanted sedge survival was greatest at more than 80% after 2 years. Although survival was not affected by surface roughness or acrotelm application, sedge growth and final cover was greater at the middle and depressional micro-topographic positions where moisture availability is expected to be greater. While sedges are known to tolerate flooded conditions (Konalova et al. 1993), they also appeared to tolerate the dryer conditions on mound tops, thereby showing greater niche breadth than *L. groenlandicum* and *P. mariana*.

Recovery from Acrotelm Application

While we expected that the addition of acrotelm would increase species richness, consistent with other studies using this technique (Quinty and Rochefort 2003, Cobbaert et al. 2004, Hope et al. 2005, Graf and Rochefort 2008, 2010), we were surprised that the indicator analysis revealed no species were associated with the acrotelm addition in the current study. Instead, acrotelm addition unexpectedly decreased many naturally re-colonizing species, including both *Populus* species, several bryophytes (mosses and lichens), and two shrubs (Table 3). This is particularly perplexing given that the acrotelm used showed an abundance of vascular plant and bryophyte propagules in the greenhouse.

Given that acrotelm was applied in a thin layer (~2 cm), this material may have dried out despite the application of straw to prevent this. Dried plant propagules would likely expire (Bowman et al. 1986) and also inhibit the germination of vascular seeds and bryophyte spores (Salonen and Laaksonen 1994). Our application depth of ~2 cm appears to be within the range of Quinty and Rochefort (2003), who recommended a "thin" layer (1–5 cm) of acrotelm be applied to produce a more or less uniform cover over the original peat soil. However, Quinty and Rochefort (2003) recommended using a rotary cultivator to prepare donor material for collection, and a manure spreader for greater fragmentation and distribution of material. Our excavator deposited fewer but larger intact acrotelm pieces. Finally, soil chemistry may have adversely affected moss establishment, together with the transplanted species. About 40% of the surface soil on the well-pad undergoing revegetation was mineral rather than organic, and soil pH was generally greater on the pad than in the surrounding peatland (Shunina 2014), both of which could have impeded moss establishment (Bragazza and Gerdtol 1996). Where resulting pH, salinity, and mineral balances may preclude use of some species as transplants, the use of several species adapted to a wide range of moisture and chemical conditions may enhance resilience during revegetation efforts, making them important to consider for peatland revegetation.

While acrotelm application was not directly beneficial in enhancing vegetation re-establishment, this material

appeared to promote the survival and growth of tree and shrub transplants. *Picea mariana* may have benefited from the root-insulation effects of added organic matter (Lambers et al. 1998), and the associated conservation of soil moisture (Caisse et al. 2008). Viereck and Johnston (1990) noted *P. mariana* performed well on heavy peat underlain by decayed wood. Although we found no relationship between *P. mariana* survival and soil moisture (data not shown), the limited scope of our environmental data did not allow us to assign causation. Moreover, straw applied to both the acrotelm and non-acrotelm treatments should have led to similar insulative and moisture retention results. Alternatively, acrotelm addition could have increased nutrient supply (Mugasha et al. 1999), particularly on areas with mineral soil at the surface. The absence of lasting benefits from acrotelm application for *P. mariana* and *L. groenlandicum* survival and growth might be attributed to the thin nature of the added material. Unless longer-term benefits are observed, the cost of acrotelm application does not appear to be warranted as a soil amendment, but may be justified as a source of live plant propagules in conditions other than those presented here.

Conclusions

While there was no difference in transplanted species' survival and growth overall between rough and smooth treatments, variation in survival and growth among microtopographic planting positions within the rough treatment leaves open the likelihood that surface roughness may be an effective management strategy to mitigate ecological risk to transplants from a fluctuating water table. In this study, providing a varied microtopography increased the probability of planted species occurring on an acceptable niche microsite, with woody species surviving and growing better on mound tops. Using this approach, although species characteristics may be fixed, the plant community's resiliency to seasonal or year-to-year moisture variations can be enhanced. The same concept applies to improving richness and diversity of naturally colonizing species.

There was little evidence that application of acrotelm fragments was a fruitful strategy for revegetating this well pad with peatland vegetation in the short-term. We could not rule out however, that factors unique to the study site, such as mineral substrate, or our acrotelm application method, may have limited the effectiveness of this treatment. Given that others have had good success with this method, we believe further testing is warranted on a larger number of well sites and over the long-term to clearly establish when, where, and how live acrotelm may benefit vegetation recovery. While transplants established and directly contributed to revegetation of this reclaimed site, the slow recovery of other species highlights the need to set realistic time frames for vegetation colonization and expansion, as well as maintain long-term monitoring

programs to further refine reclamation guidelines following the removal of infrastructure such as well pads in the boreal forest.

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